



Introduction to Azure Virtual Desktop

Azure Virtual Desktop – how it works and is it a good fit for your organization

John Kelbley - Director
Azure Advanced Migration
johnkel@microsoft.com

Shawn Meyer – Principal CSA
Microsoft Federal
shawn.meyer@microsoft.com



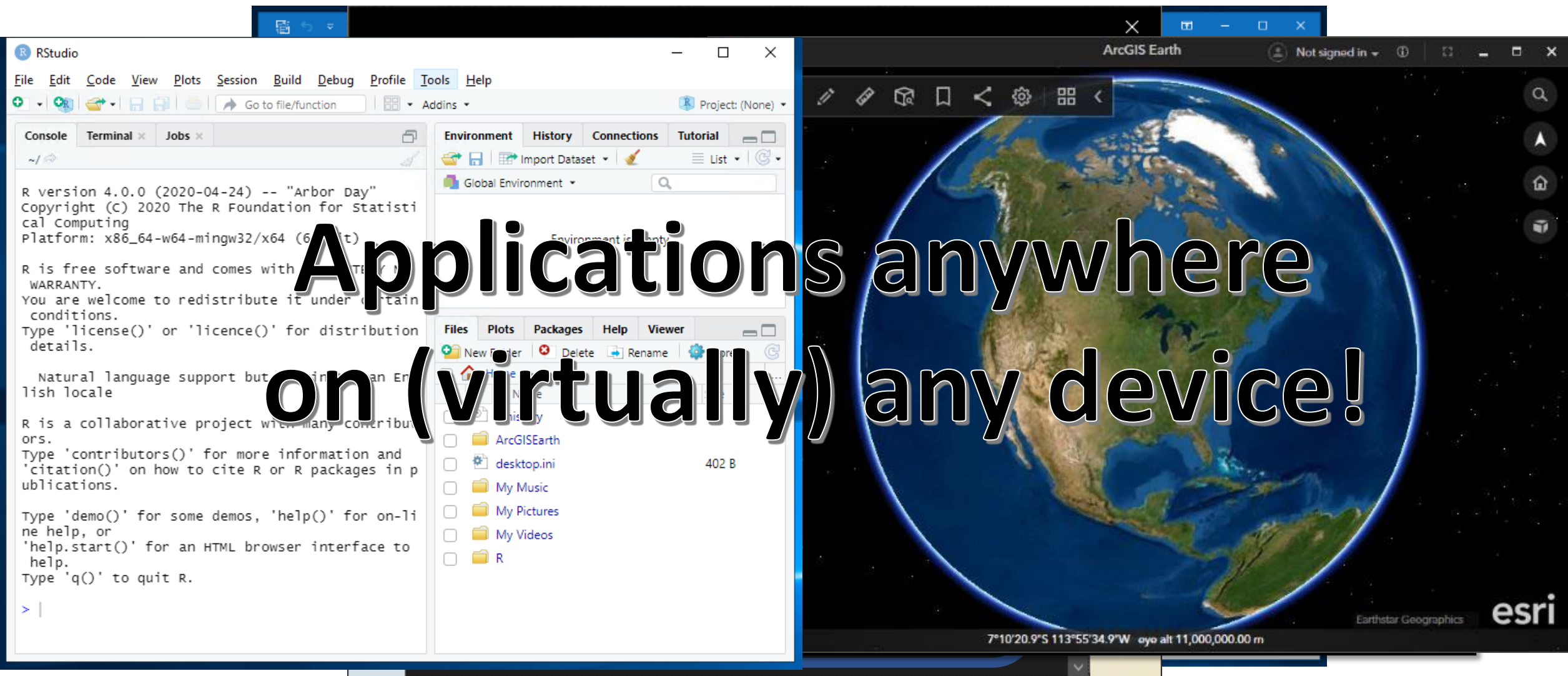
Cloud-based Solutions Provide Flexibility to Meet Evolving Demand

Azure Virtual Desktop (AVD) is the Foundation

- AVD can securely extend access to applications for Internet connected users:
- Builds upon Microsoft licensing to reduce or (in many cases) eliminate VDI-related licensing costs
- Same frontend security and telemetry infrastructure used for Office 365
 - Desktops and applications publish via Entra ID
 - Integrates with Entra ID or 3rd party MFA (including smart cards) and monitoring
 - FedRAMP High accreditation for Azure Global and Azure Government
- Works with ubiquitous applications – Office, Adobe, SAP, Esri,... as well as less common tools (subject to 3rd party licensing / activation requirements)
 - Enables centralizing / consolidating specialized resources - costly modeling / single user workstations can be shared between users and shutdown at night

What Can You Do with Azure Virtual Desktop?

**Applications anywhere
on (virtually) any device!**



The image displays two application windows side-by-side, illustrating the capabilities of Azure Virtual Desktop. On the left is the RStudio interface, showing the R console with version 4.0.0 information and a file explorer on the right. On the right is the ArcGIS Earth application, displaying a 3D globe of the Earth. The text 'Applications anywhere on (virtually) any device!' is overlaid in large, bold, white letters with a black outline, emphasizing the flexibility of the virtual desktop environment.

Prerequisites: Licensing

Many customers are already eligible for AVD:



Client - Customers are eligible to access Windows 10 / 11 single and multi session Windows with Azure Virtual Desktop (AVD) if they have one of the following licenses*:

- Microsoft 365 E3/E5 (G3/G5)
- Microsoft 365 A3/A5/Student Use Benefits
- Microsoft 365 F3
- Microsoft 365 Business
- Windows Enterprise E3/E5
- Windows Education A3/A5
- Windows VDA per user



Server - Customers are eligible to access Server workloads with Azure Virtual Desktop (AVD) if they have one of the following licenses:

- RDS CAL license with active Software Assurance (SA)
- Take advantage of options such as one-year or three-year Azure Reserved Virtual Machine Instances, which can save up to 72 percent versus pay-as-you-go pricing. Now with monthly payment options!



Compute - Pay only for the virtual machines (VMs), storage, and networking consumed when the users are using the service

- Take advantage of options such as one-year or three-year Azure Reserved Virtual Machine Instances, which can save up to 72 percent versus pay-as-you-go pricing. Now with monthly payment options!

For the latest information, please go to <https://azure.microsoft.com/en-us/pricing/details/virtual-desktop/>

Azure Virtual Desktop: Supported OS

VMs in customer's Azure subscription,
providing user experience

- **Windows 11 Enterprise Multi-session**
- **Windows 10 Enterprise Multi-session**
- Windows 11 Enterprise Single-Session
- Windows 10 Enterprise Single-Session
- Windows Server 2022
- Windows Server 2019
- Windows Server 2016



The Value of Multisession Windows Enterprise: Pooled Compared to Personal Deployment

How many fish tanks do you want to pay to maintain?



Think about per VM infrastructure including security stack, log collection, OS resources
Fewer VMs can significantly reduce cost

Azure Virtual Desktop

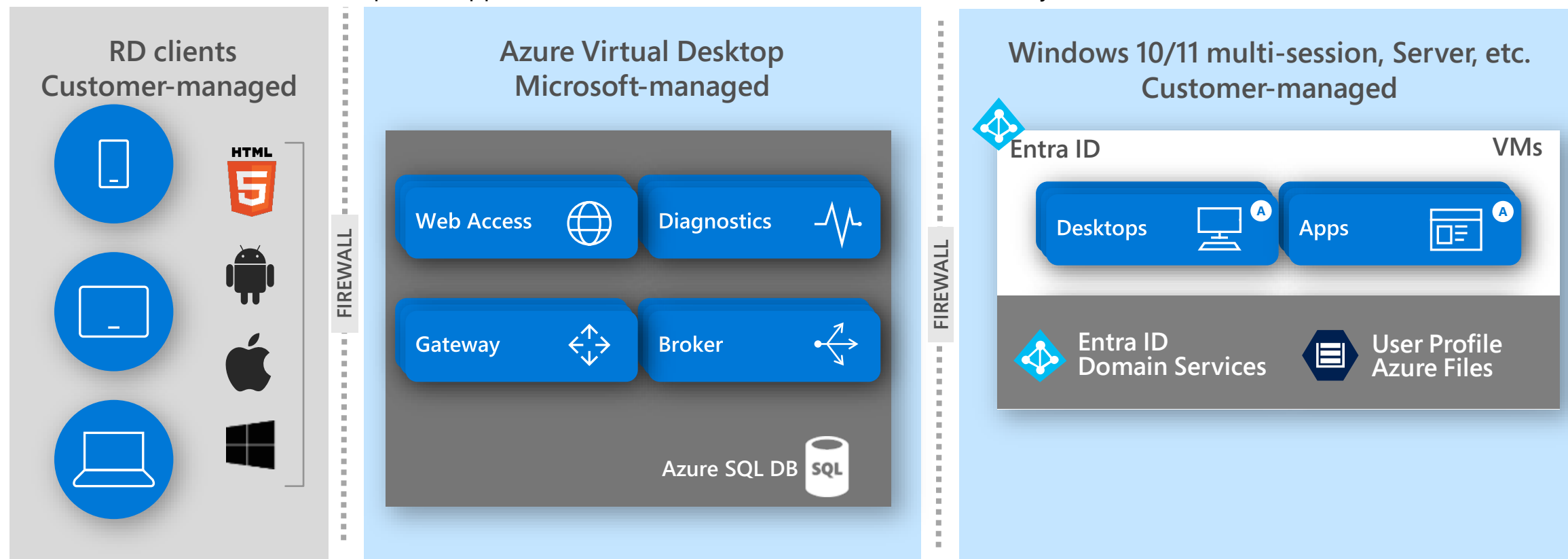
Consists of: **Azure service** to manage connections between **RD clients** and **Windows VMs** in Azure (e.g. Windows multi-session)

IT admins can:

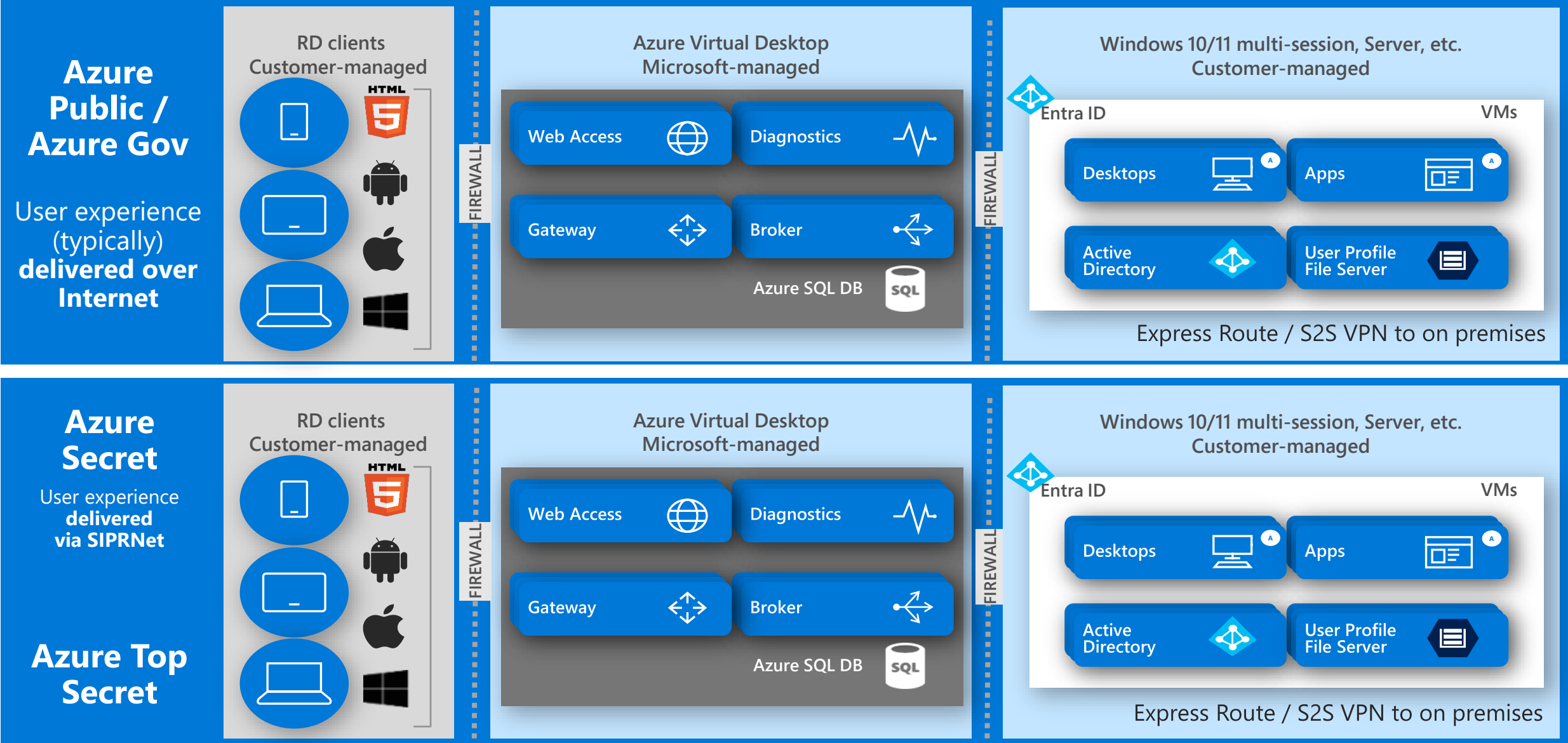
- Publish remote desktops and apps to end users from pools of single or multi-session Windows VMs in Azure
- Manage and troubleshoot connections between RD Clients and Windows virtual machines

End users can:

- Connect to Windows desktops and applications from their favorite client device from anywhere on the internet



User Experience Access





Demonstration: Azure Virtual Desktop User Experience & Administration

John Kelbley

johnkel@microsoft.com



How the NYC Department of Environmental Protection connects employees with Azure Virtual Desktop

[Microsoft Cloud Blog
https://cloudblogs.microsoft.com/industry-
blog/microsoft-in-
business/government/2020/06/10/how-the-nyc-
department-of-environmental-protection-connects-
employees-with-windows-virtual-desktop/](https://cloudblogs.microsoft.com/industry-blog/microsoft-in-business/government/2020/06/10/how-the-nyc-department-of-environmental-protection-connects-employees-with-windows-virtual-desktop/)

"Azure gave us the flexibility to be a more resilient business and maintain business continuity"

"We see this as a huge opportunity for us as an organization to move towards our digital transformation"

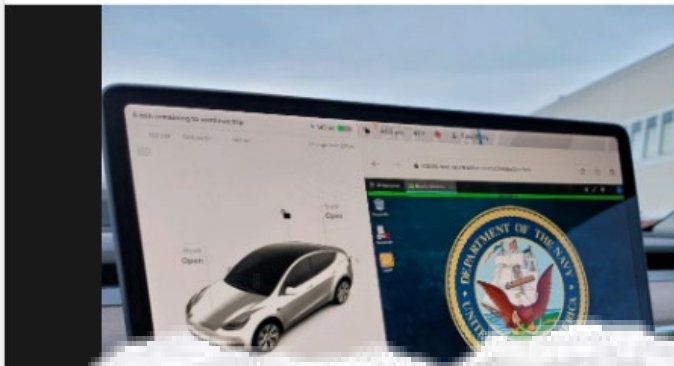
*"...enabled the agency to develop a pool of talents and enough knowledge to configure and deploy cloud technologies such as **Azure Virtual Desktop** quickly and efficiently"*



Azure Virtual Desktop Enables Secure, Anywhere Access



- Deployed and in use across the DoD (Navy Flankspeed, Army, Air Force, Coast Guard, others)
- Builds upon existing licensing investment
 - No need for additional 3rd party software
 - Over the Internet to NIPR for IL4/IL5 (AVD Available on SIPR as well) – **no need for GFE!**
 - Optimized routing via world-wide Microsoft network to Azure Government
- Capabilities vary by deployment
 - Scales dynamically to meet demand
 - Multi-Monitor
 - Clients for Windows, Mac, iOS, Android, web
 - CAC-enabled for Windows / Mac

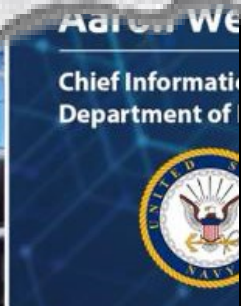


AVD in the Department of Defense

Navy used threat of cyber vulnerability to expand VDI
“The DoN expanded its virtual desktop interface to 110,000 users from 25,000 in less than a week...”



[Navy used threat of cyber vulnerability to expand VDI \(federalnewsnetwork.com\)](https://www.federalnewsnetwork.com)



digital
transformation and AVD

- Coast Guard delivers [Manta Virtual Desktop](#)

Thank You!

Learn more about Azure Virtual Desktop – Additional sessions Today!

- 10:30AM – 11:00AM
- **12:45PM – 1:30PM**
- 1:45PM – 2:30PM
- 3:00PM – 3:45PM

Introduction to Azure Virtual Desktop

Azure Virtual Desktop: Real World Use Cases

Deploying and Managing AVD

Azure Arc: Optimizing Server Management

- 5:30PM – 6:30PM

Happy Hour sponsored by



John Kelbley - Director
Azure Advanced Migration
johnkel@microsoft.com

Shawn Meyer – Principal CSA
Microsoft Federal
shawn.meyer@microsoft.com



FEDERAL AZURE SUMMIT AGENDA

JAN 16
2025

8:30 - 9:00

Registration and Welcome Coffee

9:00 - 9:15

Opening Remarks by [Javier Vasquez](#)

9:15 - 9:45

Keynote: Future of Hybrid Cloud by [Jason Payne](#)

9:45 - 10:15

Adaptive Cloud: Extending Cloud Capabilities to the Edge by [Eric Sanchez](#) / [Joe Fox](#)

10:15 - 10:30

Morning Break

10:30 - 11:00

Introduction to Azure Virtual Desktop (AVD) by [John Kelbley](#) / [Shawn Meyer](#)

11:00 - 11:45

Introduction to Azure VMWare Solution by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

11:45 - 12:45

Lunch Sponsored by



12:45 - 1:30

Session 1A: Real-World Use Cases of Azure Virtual Desktop by [John Kelbley](#) / [Shawn Meyer](#)

Session 1B: Real-World Use Cases of Azure Edge Devices by [Rich Shillington](#) / [Jason Benson](#)

1:45 - 2:30

Session 2A: Deploying and Managing AVD by [Shawn Meyer](#)

Session 2B: Real-World Use Cases of Azure VMWare Solutions
by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

2:30 - 2:45

Afternoon Break

3:00 - 3:45

Session 3A: Azure Arc – Optimizing Server Management by [John Kelbley](#)

Session 3B: Deploying & Managing Azure VMWare Solutions
by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

4:00 - 4:45

Fed Chief Architect Panel: The Future of Cloud and Edge Computing by [Jason Payne](#) / [Chief Architects](#)

4:45 - 5:00

Closing Remarks and Networking by [Javier Vasquez](#)

5:30 - 6:30

Happy Hour Sponsored by



Please visit <https://aka.ms/FedAzSummit/home> for more information

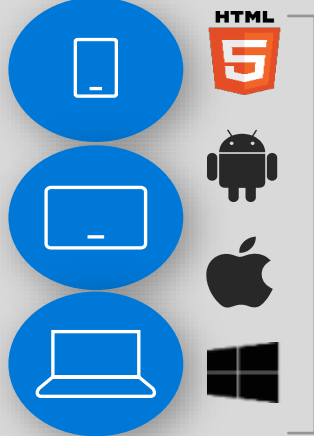
High-Level Architecture: AVD Compared to Azure Stack HCI

Azure Virtual Desktop

**Azure
Public /
Azure Gov**

User experience
(typically)
**delivered over
Internet**

RD clients
Customer-managed



FIREWALL

Azure Virtual Desktop
Microsoft-managed

Web Access



Diagnostics



Gateway



Broker



Azure SQL DB



FIREWALL

Windows 7 / 10 / 11 single or 10 / 11 multi-session, Server
Customer-managed

VMs

Desktops

Apps

Entra Domain
Services

User Profile
File Server

Express Route / S2S VPN to on premises

(on Azure Stack HCI)

Desktops

Apps

Entra Domain
Services

User Profile
File Server

RDP ShortPath for Public Networks
uses Customer network
after session started

(low latency!)

AVD VMs run inside
customer Azure subscription
Gateway /

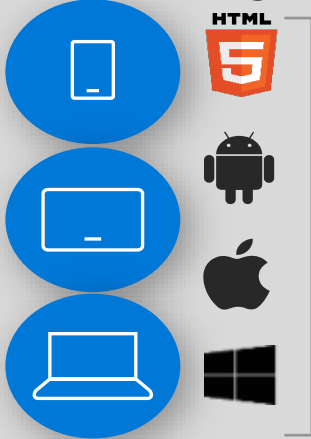
High-Level Architecture: AVD Compared to Windows 365

Azure Virtual Desktop

Azure
Public /
Azure Gov

User experience
(typically)
delivered over
Internet

RD clients
Customer-managed



FIREWALL

Azure Virtual Desktop
Microsoft-managed

Web Access



Diagnostics



Gateway



Broker



Azure SQL DB



FIREWALL

Windows 7 / 10 / 11 single or 10 / 11 multi-session, Server
Customer-managed



Entra ID

Desktops

Apps

Entra Domain
Services



User Profile
File Server



Express Route / S2S VPN to on premises

AVD VMs run inside
customer Azure subscription

Windows 365

W365 is connected to customer Azure
(runs in Microsoft subscription)

FIREWALL



Entra ID

Windows 10 / 11 single-session VMs



MEM

Desktops

The benefits of Nerdio



Cost insights &
reporting



Image lifecycle
management



Environment
rightsizing



DR & capacity
mitigation



Application
management



Policy
management

AI assistance



HCI stack
support



Modern endpoint
management



End-user self
service